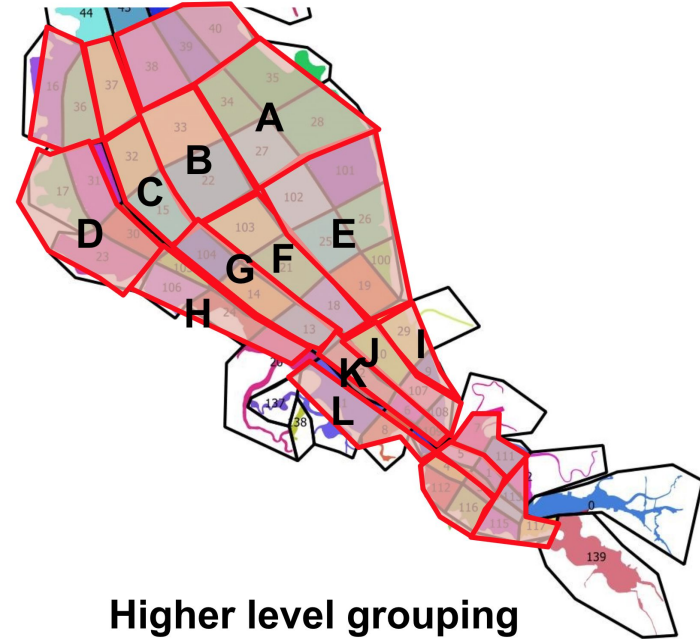
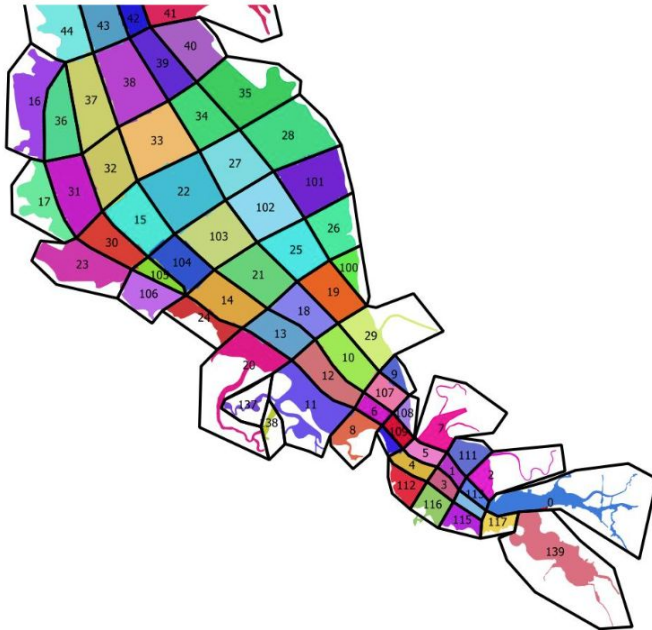


- Mass Balances: subembayment and sub-subembayment
- Transport pathways
- Nutrient-loss zones
- Phytoplankton production zones



Where does this work fit in with FY20 work?

1. To understand how the model is working, and whether the predictions make sense, we need to do this level of mechanistic exploration, striking a balance between high-resolution enough that we're not aggregating together regions having distinct forcings and biogeochemical responses, but aggregated enough that we can compare sub-regional results with conceptual models, rates from other systems, etc.
2. Important for on-going knowledge-transfer and capacity building related to biogeochemical modeling.
3. What we learn here will help inform sensitivity analysis work, and is needed to inform model calibration for WY2013 (v2.0) and WY2017 (v1.0). The scripts and visuals that we develop will be used for updated calibrations of the high-res model, and used to examine the effects of changing parameters not just state variable concentrations but on the relative importance of rates, over space and time.
 - a. WY20 Deliverable #1: Updated/Refined model calibration for WY2013
 - b. WY20 Deliverable #2: Model calibration for WY2017
 - c. WY20 Deliverable #3: Progress-Report, Sensitivity analysis to inform calibration and priority data collection
4. WY20 Deliverable #4: Nutrient Cycling and Fate in SFB, subembayment budgets, and relative contribution from individual POTWs to ambient nutrients(x,y,z,t)

All of the questions and analyses on p3 are priorities for FY20 projects. Some of them are higher priority for mid-next-week than others. I don't know how straightforward vs time-intensive pursuing some of them will be, so I also gave them rough priorities in terms of this week vs. later.

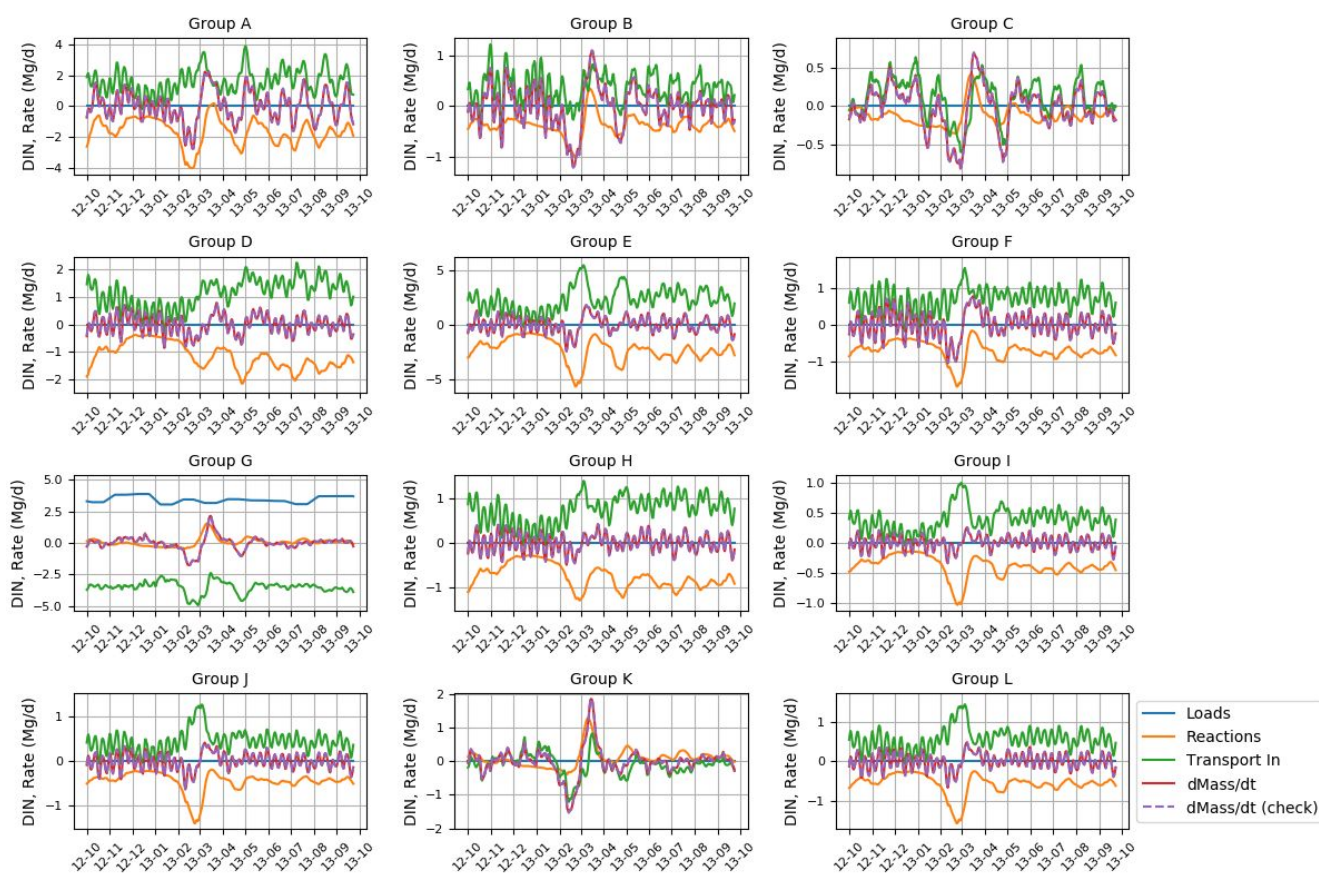
If we don't accomplish them this week, let's keep pushing on them, targeting mid-November.

******In terms of developing the code and graphics for these interpretations...For this week's push, the primary objective is completing the highest priority analyses/visuals. It definitely makes sense to develop the code, as time allows, with the assumption that we will use it again for refined calibrations with WY2013, and eventually for WY2017 and other years. So some investment toward documenting and making it flexible is worthwhile. But it doesn't need to be fully auto, and some of the clean-up and documentation could happen in subsequent weeks.

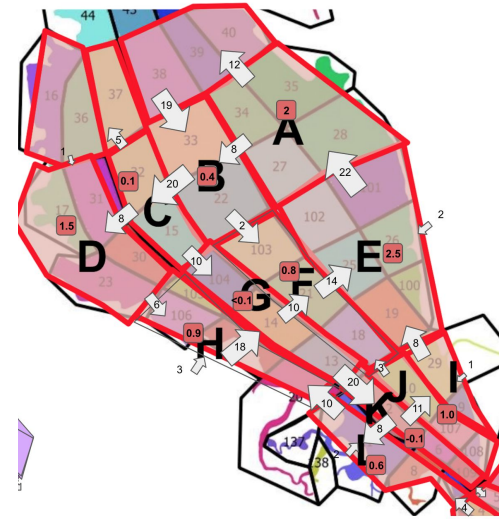
Point	Near-term priority level (goal: Oct17/18)	Questions/Goals motivating analysis/visualization
		1. What are subbayment mass balances for N?
	5	a. Focus on South Bay, and Lower South Bay. Need to consider two different boundaries for South Bay. (slide
	5	b. Primary focus on DIN
	3/4	c. Need to quantify role of phyto-N and DON+PON, ensure we're not overlooking a major term. phyto-N (see slide 9) below appears to be small most of the time (not certain we have DON+PON available to do this check).
		2. Within South Bay...[My thinking, after exploring things today is that it would be useful/informative to extend the CV work and the rough analysis below up to the Bay Bridge (i.e., so EBDA and SFPUC are inside the area being examined, as opposed to at its boundary]
	5	a. sub-subembayment mass-balances, select CVs (already have raw materials for this, AK plots). Specifically...
	5	i. Most important locations for N loss? quantify, visualize (temporal variation?)
	5	ii. Interesting/important N transport pathways? quantify, visualize (temporal variation?) [see **]
	4	b. Most important locations for phytoplankton production?
	4	i. Interesting/Important transport pathways for phyto-biomass? [see **]
	2	ii. How much of phyto-production within a CV remains within the control volume? How much exported and what are major source and sink zones?
	2	c. Characterizing N turnover times, 'flushing rates', loss rates...looking at those in terms of the magnitude of rate or transport-flux relative to mass within CV; e.g., (kg)/(kg/d)=days; or (kg/d)/(kg) = %/day).
Priorities 5(high) $\longleftrightarrow$ 1(low)		**Note: For visualizing transport would we be better off using the ~140 CVs instead of A-I? (only south bay)! bet an animation of arrow length using the 140 CVs (spring-neap filtered, daily or monthly or seasonal) will allow us to see changes in "flow" patterns over time. At lower-res (what I did below), it may be harder to see those easily, at least in terms of a presentation/animation.

Things still to resolve

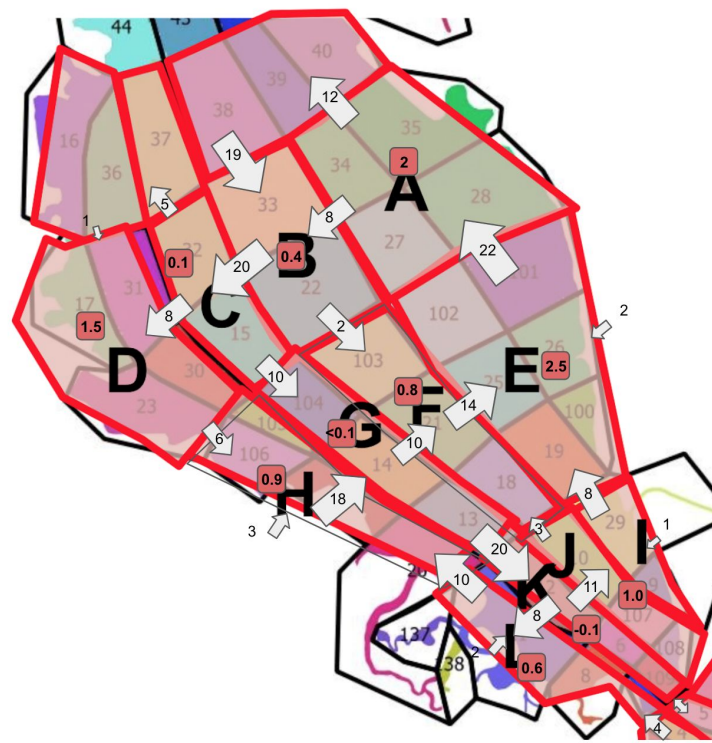
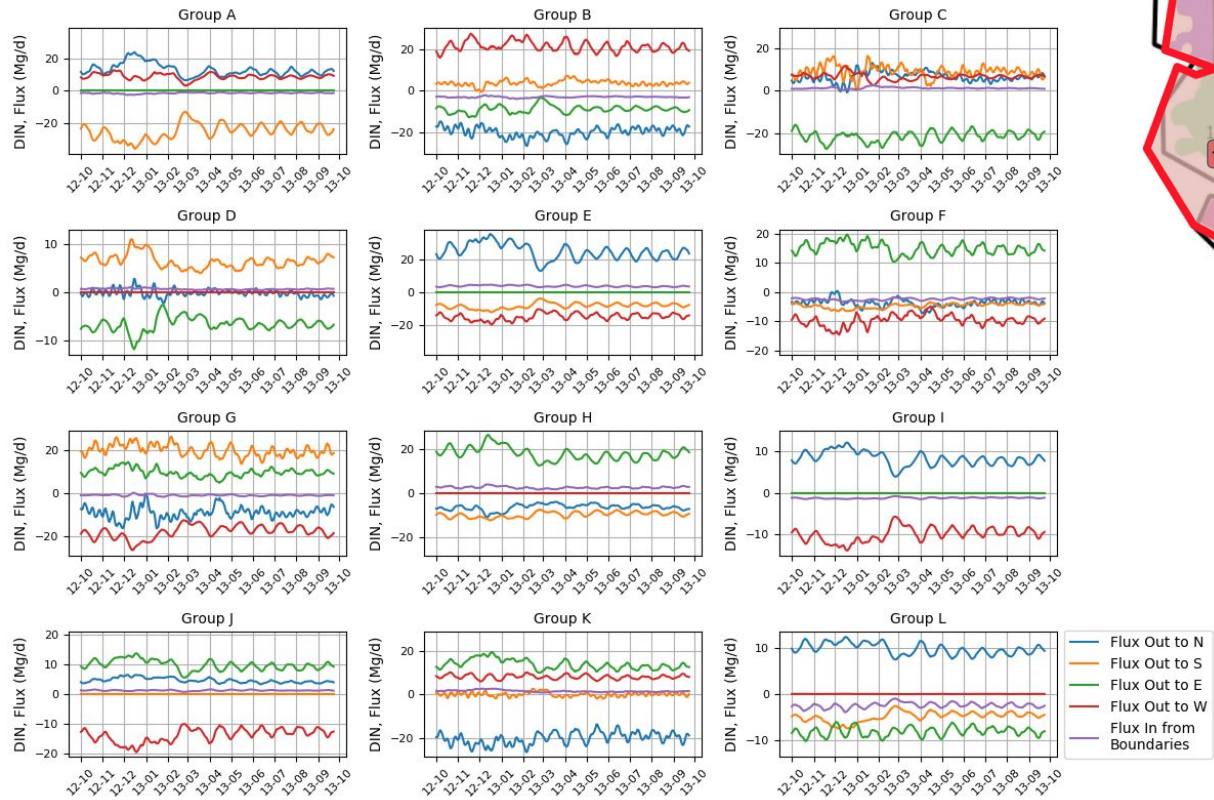
1. Do we have data on organic N (not associated with phyto/diatom; PON, DON, or both)?
2. (lower priority, something to come back to) Flux in from boundaries...couple things caught my eye... (The values are small and aren't important to mass balances. Asking more to just double-check on approach, b/c I recall this was a bit of tricky spot, make sure there isn't something going on like a missed term.
 - a. Some non-zero boundary fluxes from CVs that are in the interior. e.g.,
 - i. B: it's magnitude is comparable to E which is where Alameda Creek enters.
 - ii. C, G, K...even though the boundary flux term is small, shouldn't these be zero b/c they are interior? Also, values for G are negative.
 - b. I: values are negative.
 - c. E: relatively constant year-round, despite this being where Alameda Creek enters. Just want to confirm. This could simply be the result of us not have good / reasonable [DIN] for stormwater flows. Also, E and H are comparable magnitude, even though (i think) flows to E are much smaller than those to E.



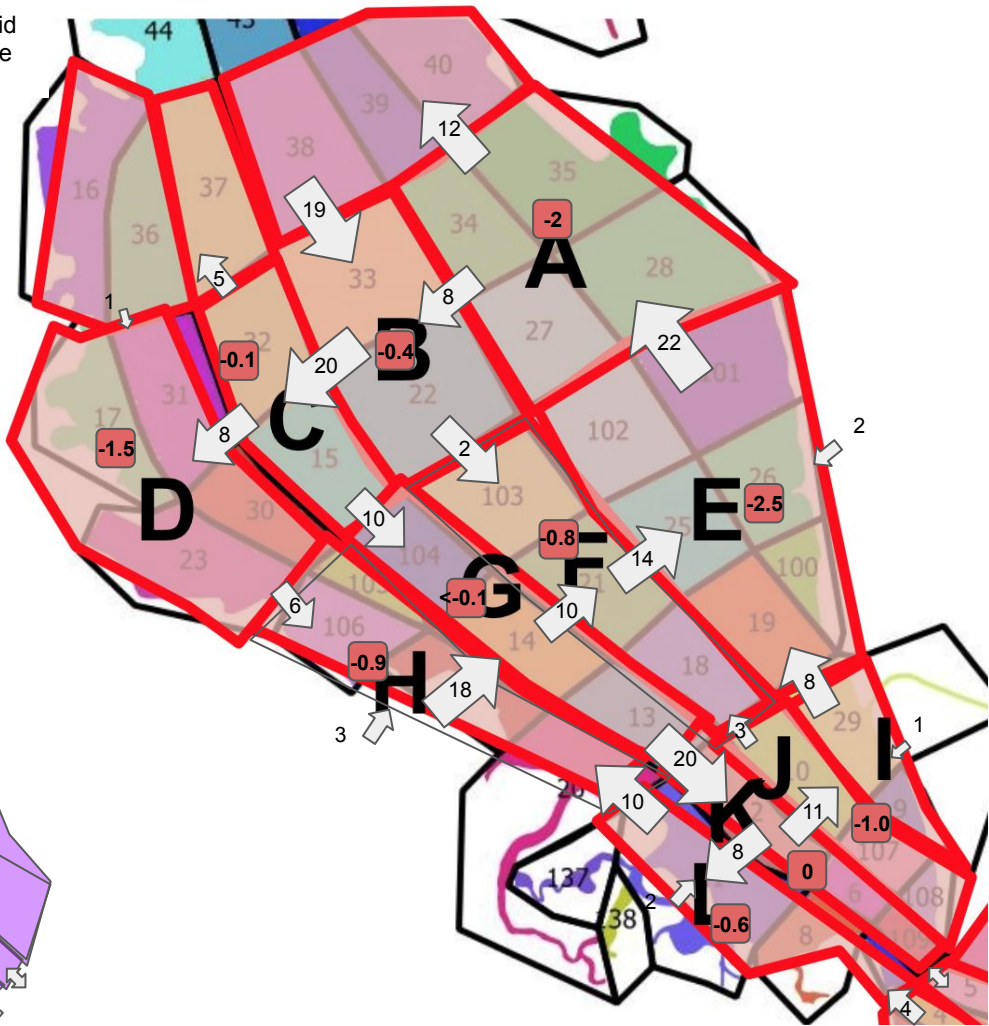
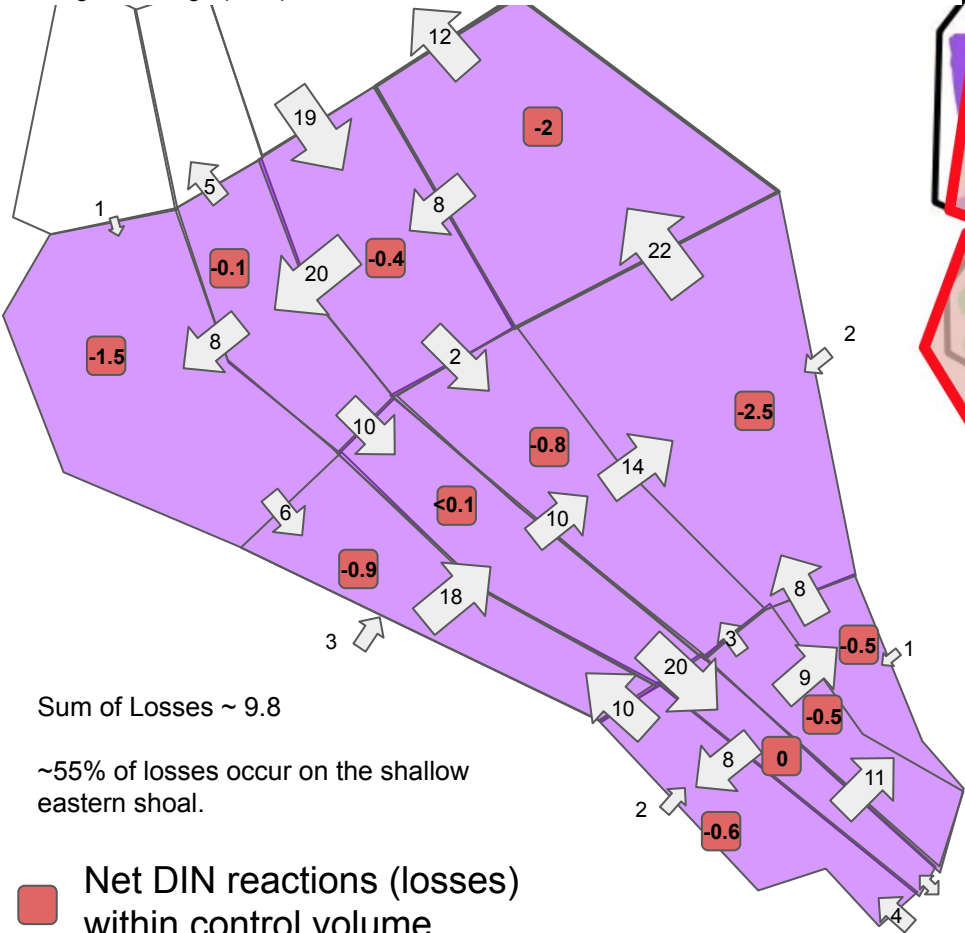
See [here](#) for monster folder(s) of plots Allie made.
Below is summer 2013 approx



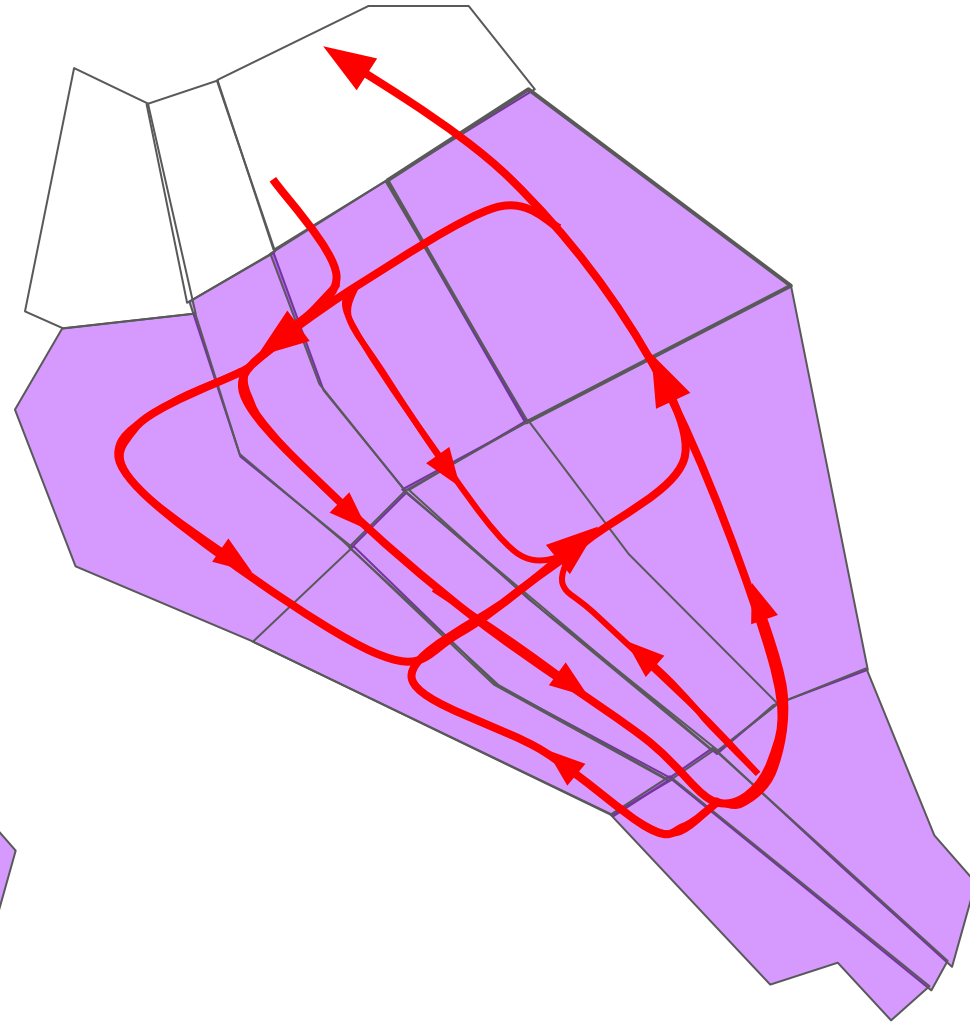
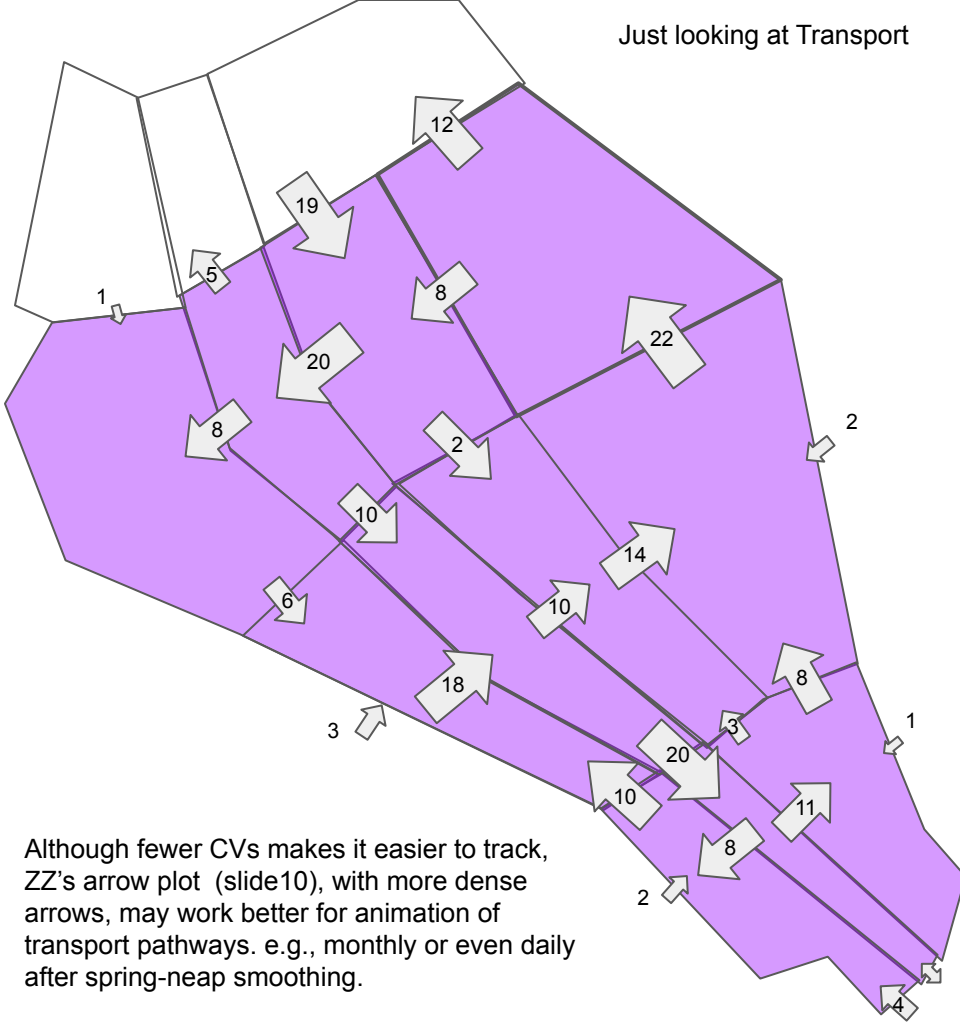
See next page for an easier to look at version on next page.



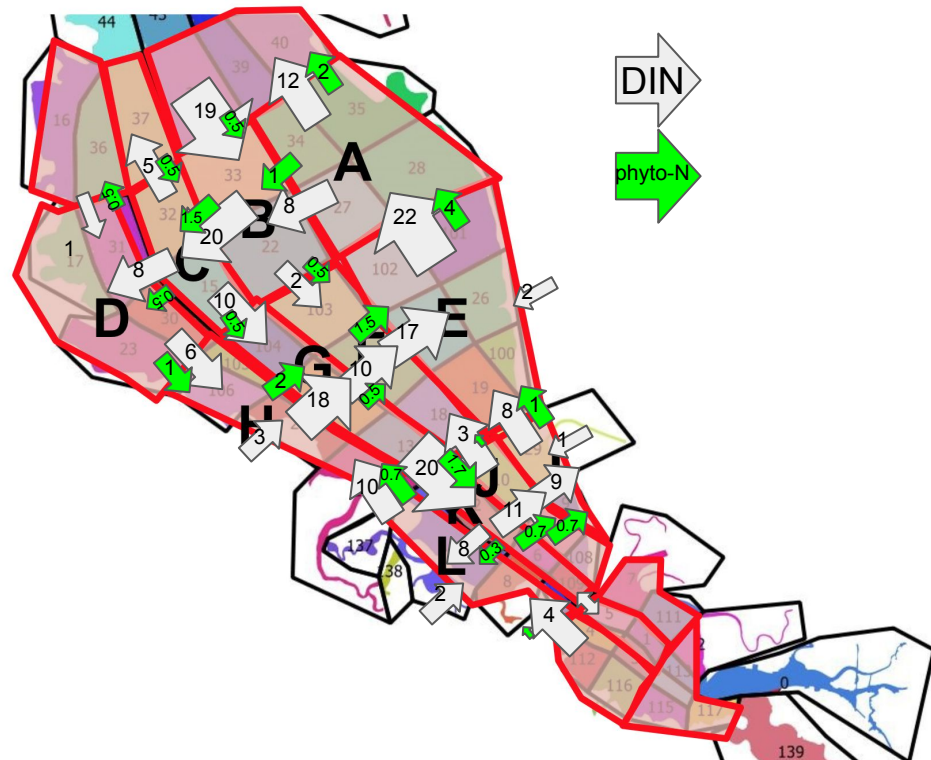
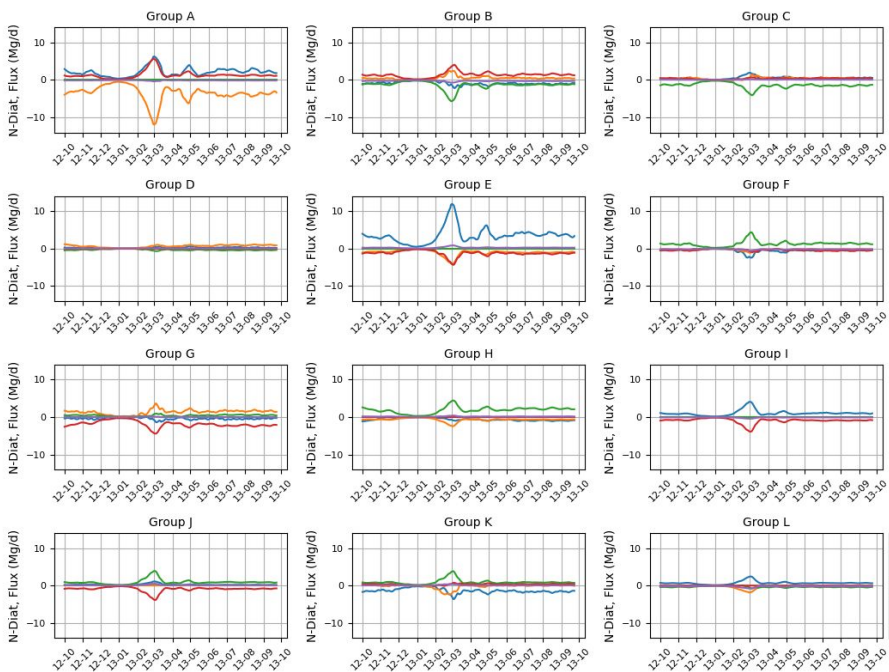
DIN fluxes and transformations. Approx values for ~summer June/July 2013. Note: I did these by eyeball from the plots. Budgets don't sum to zero, some of which could be due to change in storage (conc). Also need to look at PON+DON



Just looking at Transport

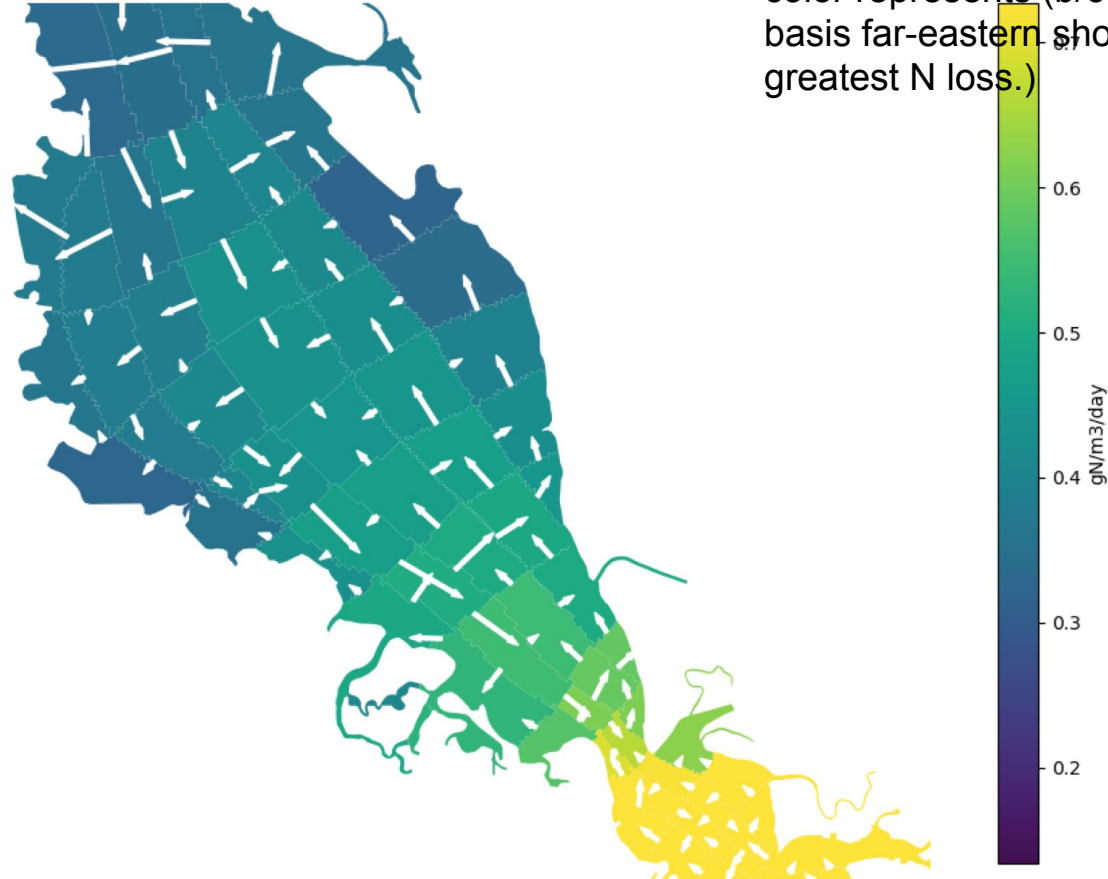


Although fewer CVs makes it easier to track, ZZ's arrow plot (slide10), with more dense arrows, may work better for animation of transport pathways. e.g., monthly or even daily after spring-neap smoothing.



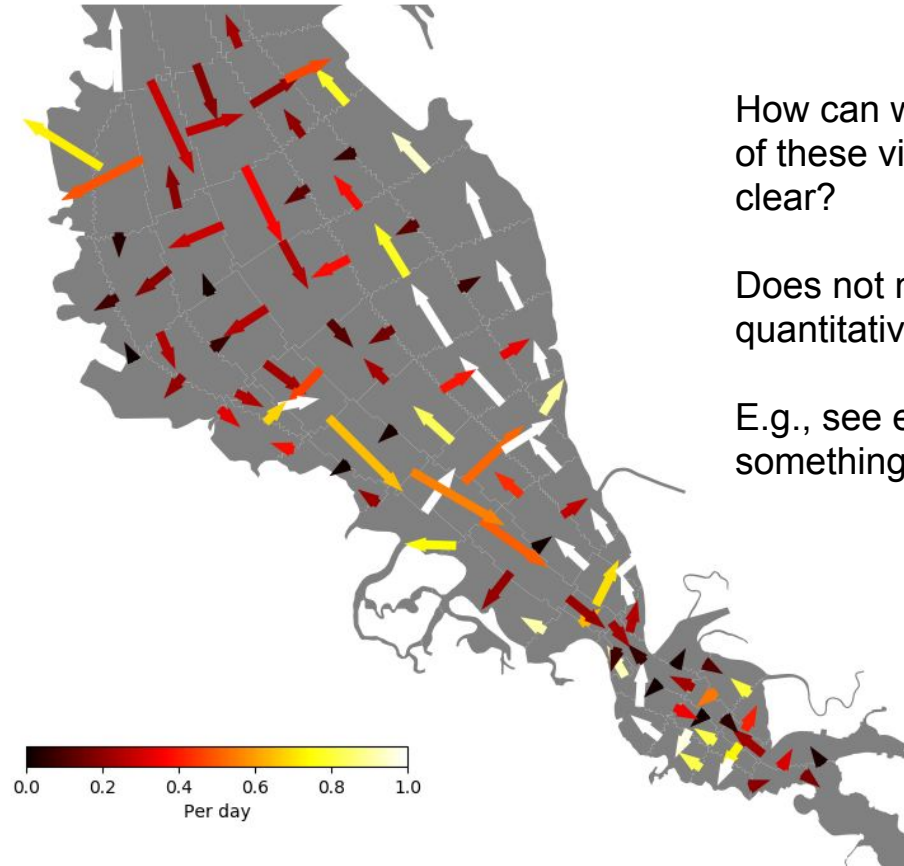
['no3', 'nh4']

Arrows represent annual-average transport fluxes. Unclear on what color represents (b/c on a mass basis far-eastern shoal is area of greatest N loss.)



2013-05-14

Example
Arrow length = Mg/d
Ignore color



How can we make an animation
of these visually-accessible /
clear?

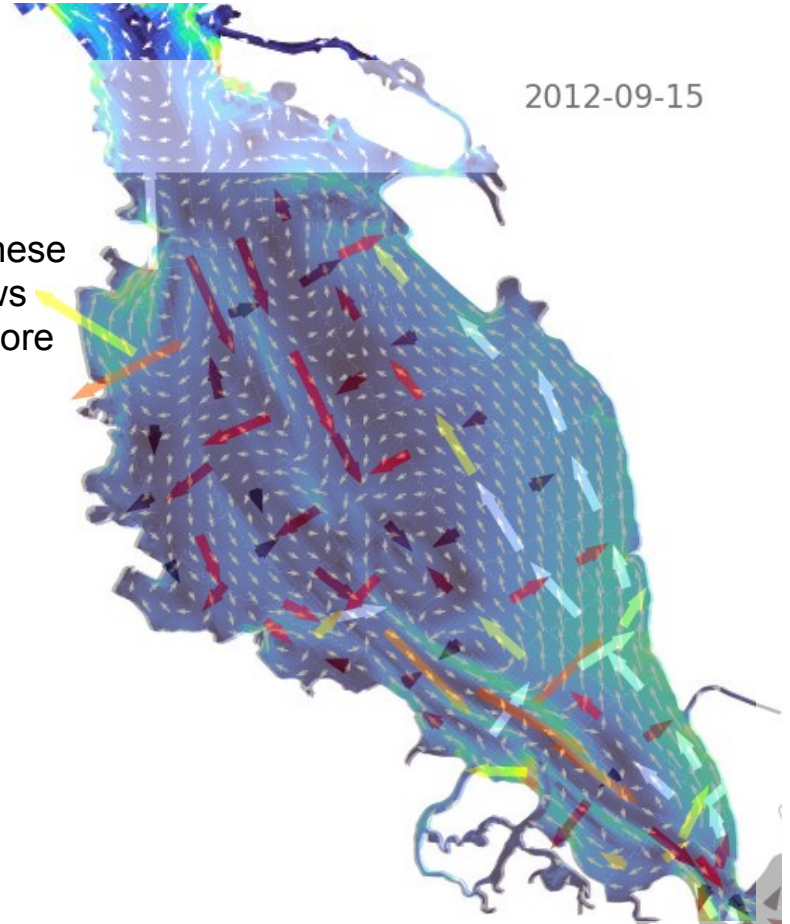
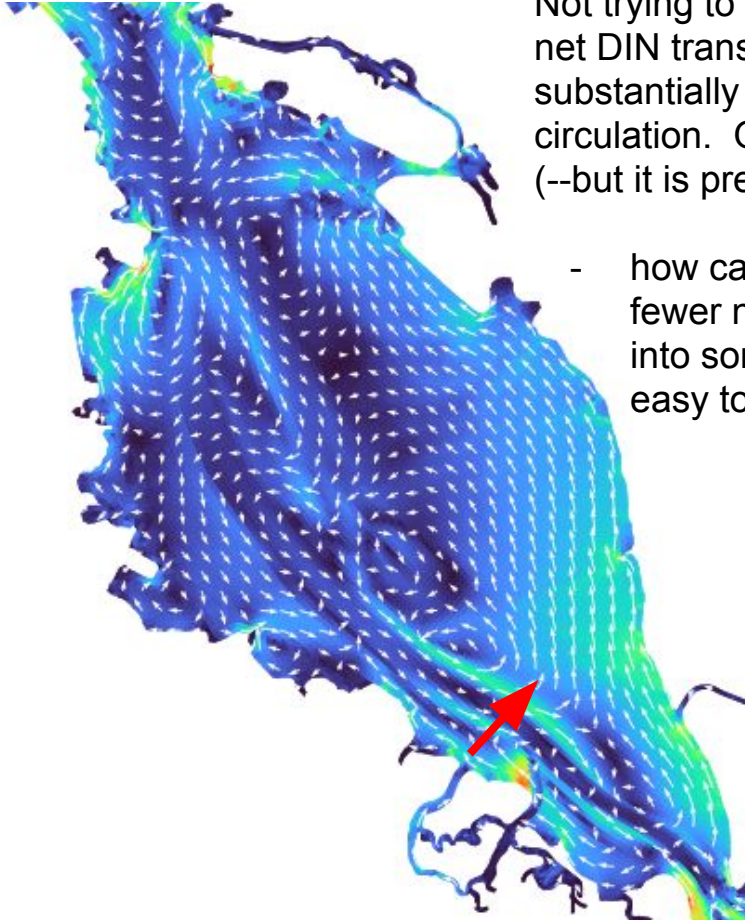
Does not need to be strictly
quantitative.

E.g., see example on next page,
something

Not trying to make case that the net DIN transport is all or substantially due to residual circulation. Oct 2012
(--but it is pretty similar)

2012-09-15

- how can we smooth these
fewer number of arrows
into something a bit more
easy to see



San Pablo Bay

Suisun Bay

“RMP-SouthBay” boundary

Approximately the boundaries between
28,27,22,15,30,23 || 35,34,33,32,31,17

WaterBoard South Bay boundary (Bay Bridge),

Approximately the boundaries between
46, 45 || 49, 47,48

Lower South Bay boundary (Dumbarton Bridge),

Approximately the northern boundary of
7,5,4,112

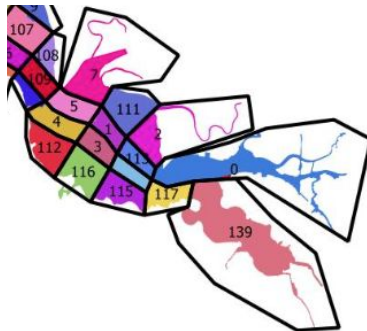
Central Bay

WB South Bay
boundary

RMP South Bay
boundary

South
Bay

Lower
South Bay



Example Graphics for Subembayment Mass Balances/Budgets

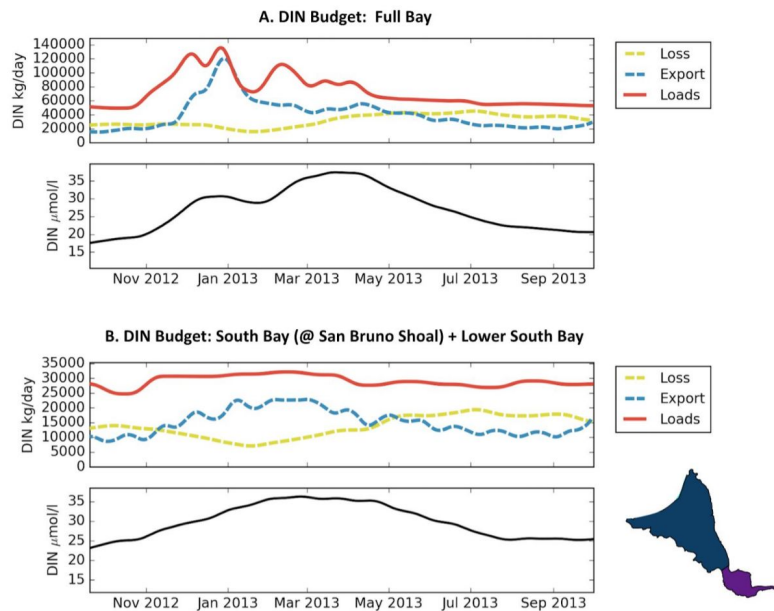


Figure 8. DIN budget for the full Bay (top panels) and South Bay + Lower South Bay (lower panels) for WY2013, based on an early version of the NMS biogeochemical model. The South Bay northern boundary used for this analysis is the San Bruno Shoal (e.g., comparable to that used in Smith and Hollibaugh, 2006). For each budget, the top panels depict DIN loads into the system, DIN losses within the system (denitrification was the loss considered), and export (computed and summed at each time step across the exiting control volume surface). Changes in DIN concentration over time within each control volume represent the changes in DIN storage (or mass) within the system, and are average concentrations within the region considered (top: full bay; bottom: South Bay and LSB). The model used for this simulation was well-calibrated for hydrodynamics but used a pared down biogeochemical model, including only nitrification and denitrification, and rates for these processes was optimized to maximize predictive skill for DIN (and DIN speciation) relative to observations over the full domain and one year simulation. The model performed reasonably well predicting biweekly-monthly DIN concentrations and speciation, providing confidence in these bulk, large-scale budgets.

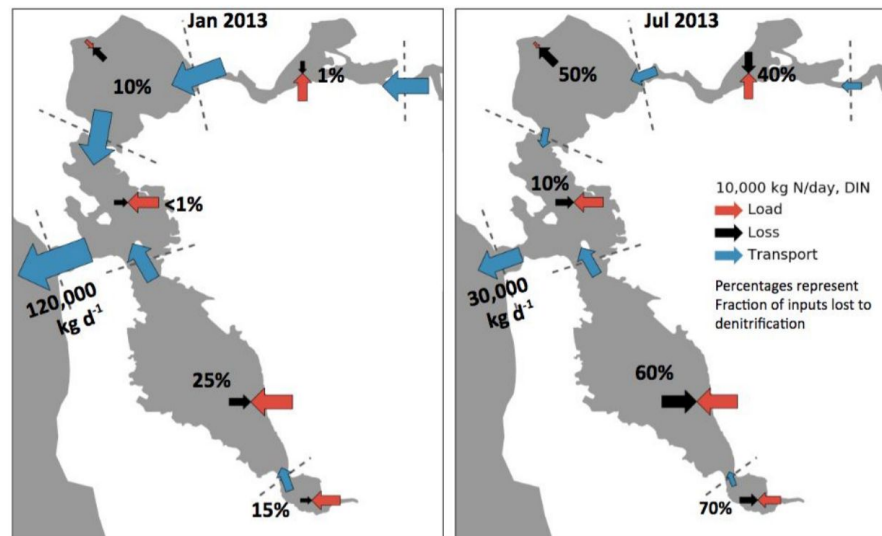
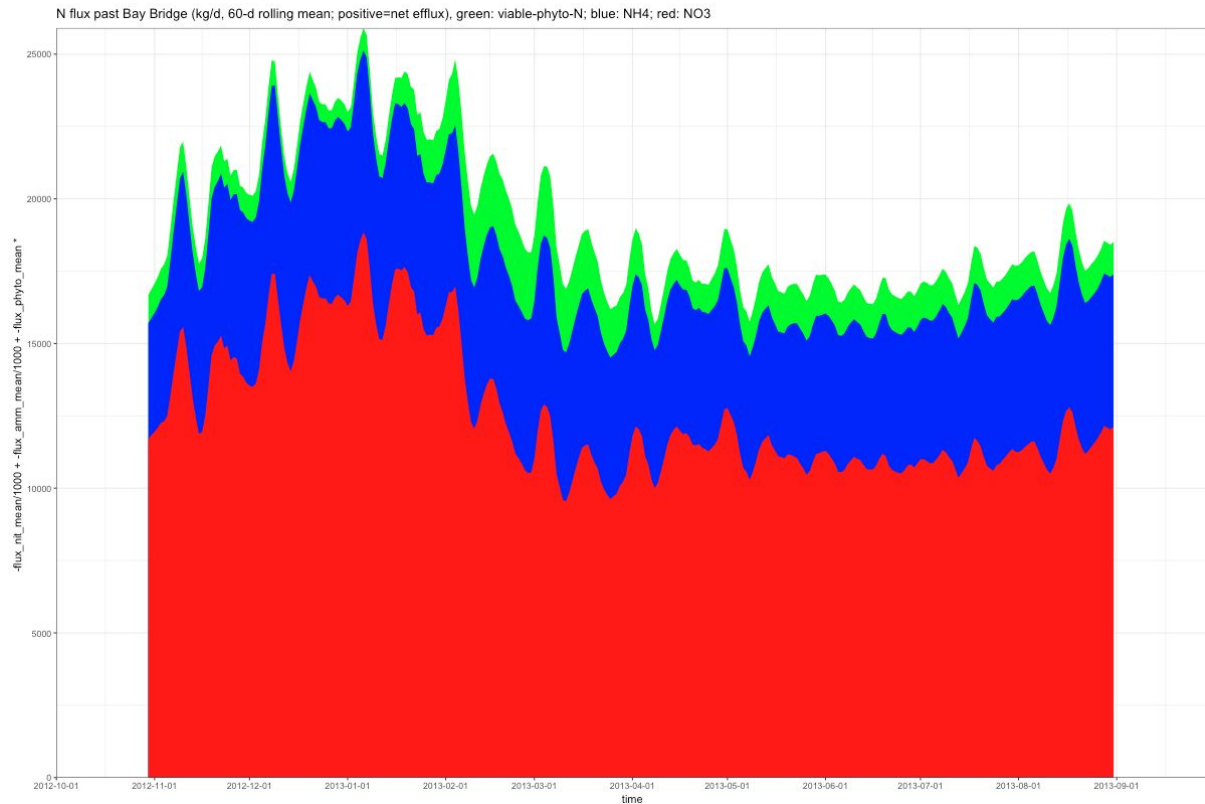


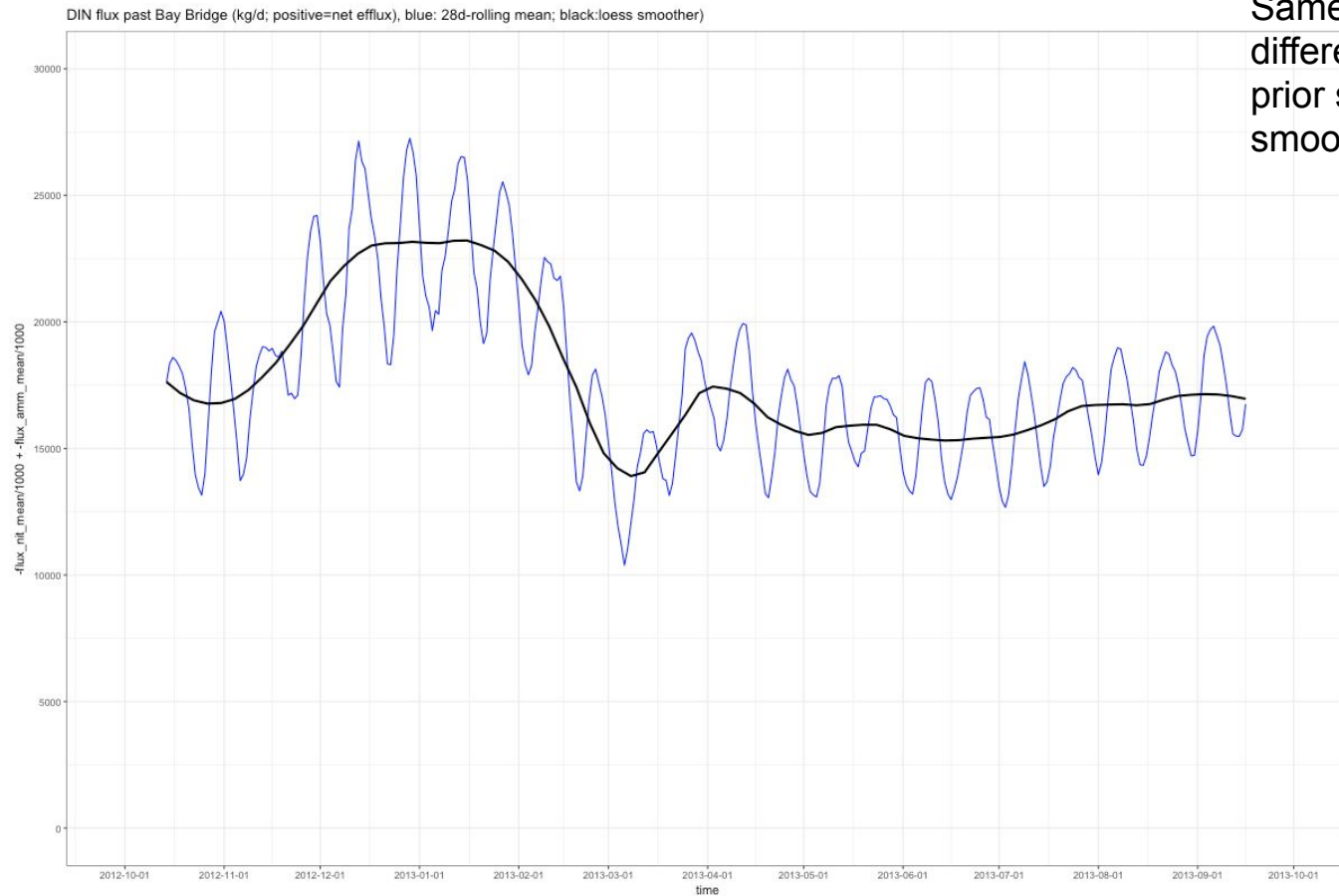
Figure 9. Monthly DIN budget for each of SFB's 5 subembayments for Jan 2013 and Jul 2013. Size of arrows are proportional to mass flux term (load, loss, transport), with arrow sizes in the legend corresponding to 10,000 kg d^{-1} . Percentages within each control subembayment denote the percentage of total DIN inputs (load + incoming transport) that was lost within the control volume. Note that South Bay northern boundary for this analysis is at the Bay Bridge, which is the regulatory boundary for South Bay. See model details in prior figure caption.

Something missing from the plots on the left is transport into the control volume from up-estuary. (wasn't needed b/c South and LSB were combined).



I made this plot a month or so ago. Needs to be checked. And also needs the terms like flux in (from LSB, when separated) direct loads to the subembayment, etc.)

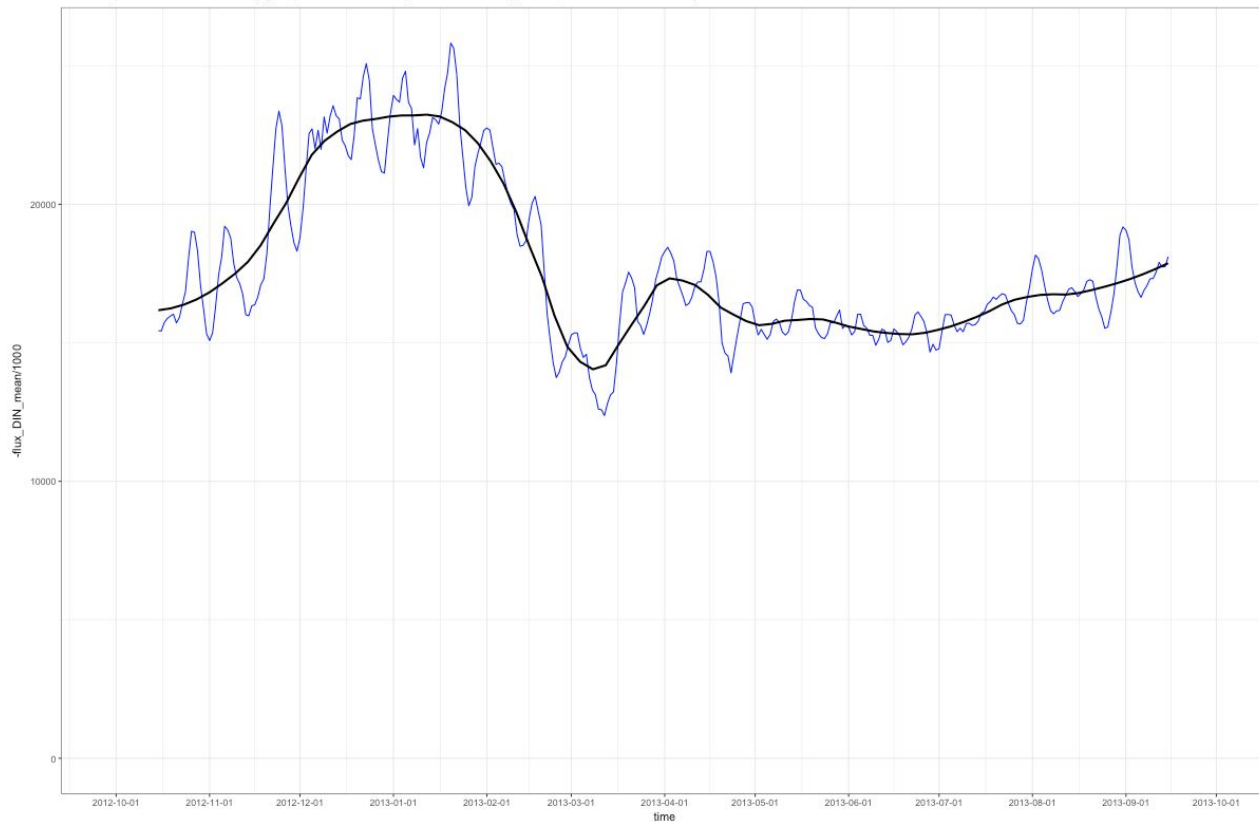
South Bay (bay bridge) mass balance



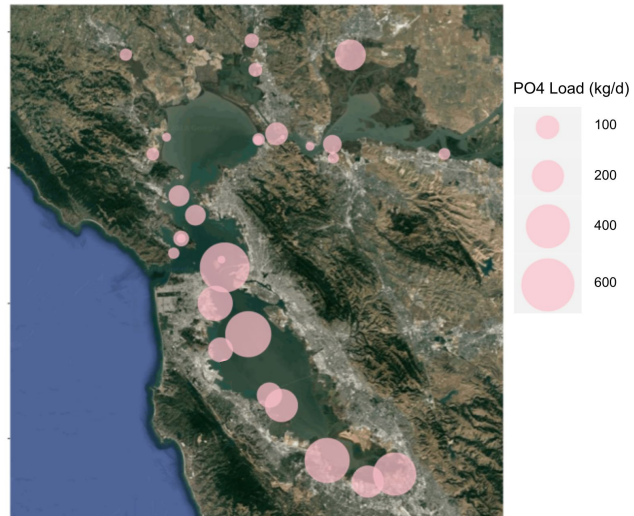
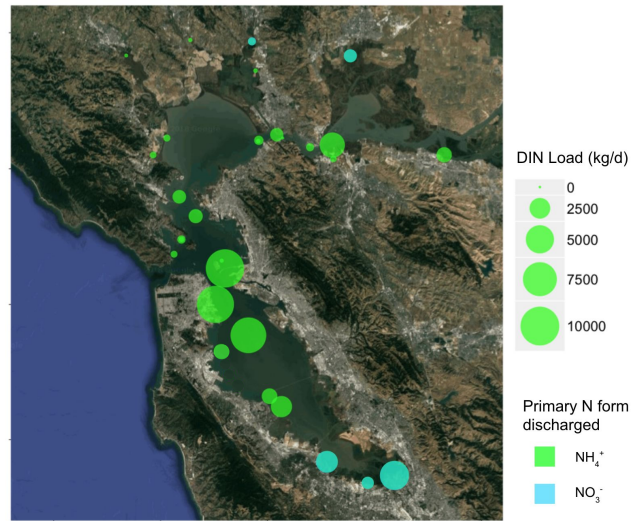
Same as above. Slightly different peak heights from prior slide may be due to smoothing

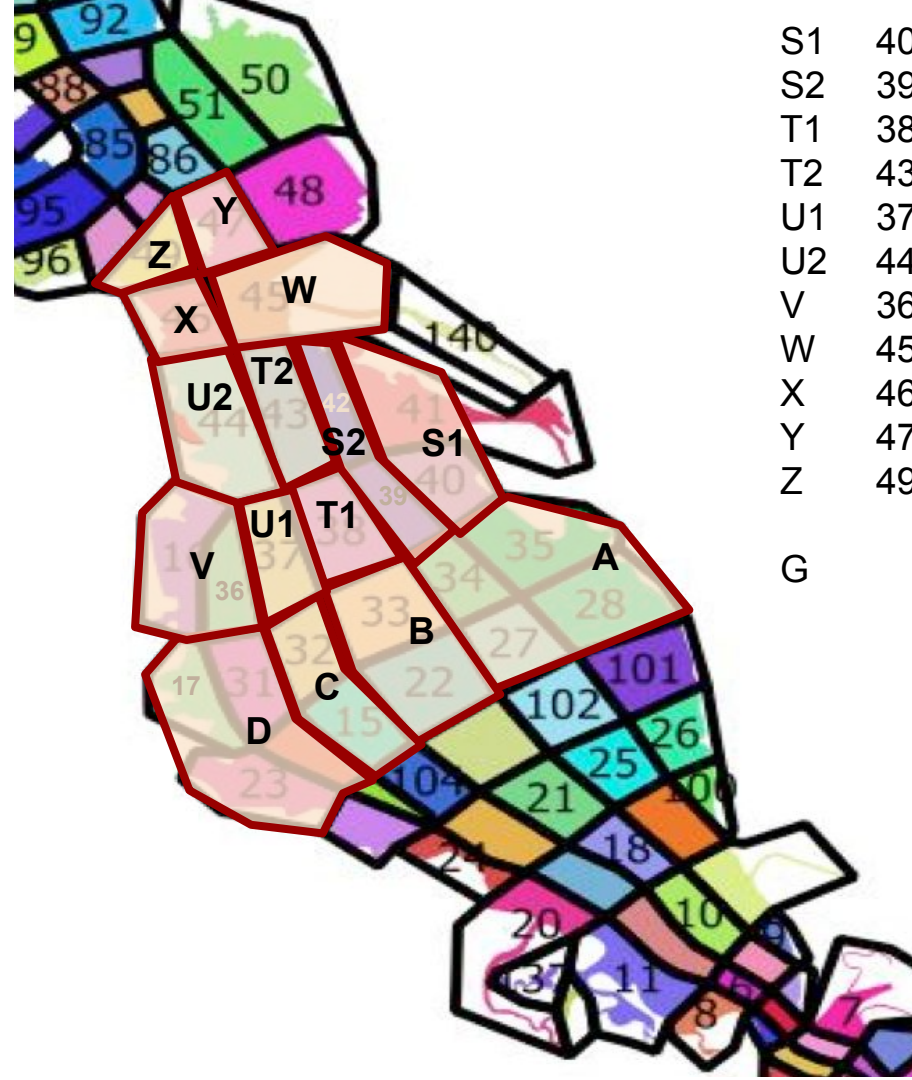
South Bay (San Bruno Shoal) mass balance

DIN flux past San Bruno Shoal (kg/d; positive=net efflux), blue: 30d-rolling mean; black:loess smoother)



Locations and sizes of WWTP discharges





S1	40,41
S2	39, 42
T1	38
T2	43
U1	37
U2	44
V	36,16
W	45
X	46
Y	47
Z	49
G	

CV labeled number and letter (S1-S2, T1-T2, U1-U2)

+7

- For now I'm suggesting that we keep them as individual CVs.

- Later we may want to combine them (either north-south pairs or east-pairs), but I'm not sure at the moment.

+8

+2

- Reason to keep separate T1-T2 and U1-U2 for now.
 - EBDA discharges in T1(38), SFPUC discharges in U2 (44).

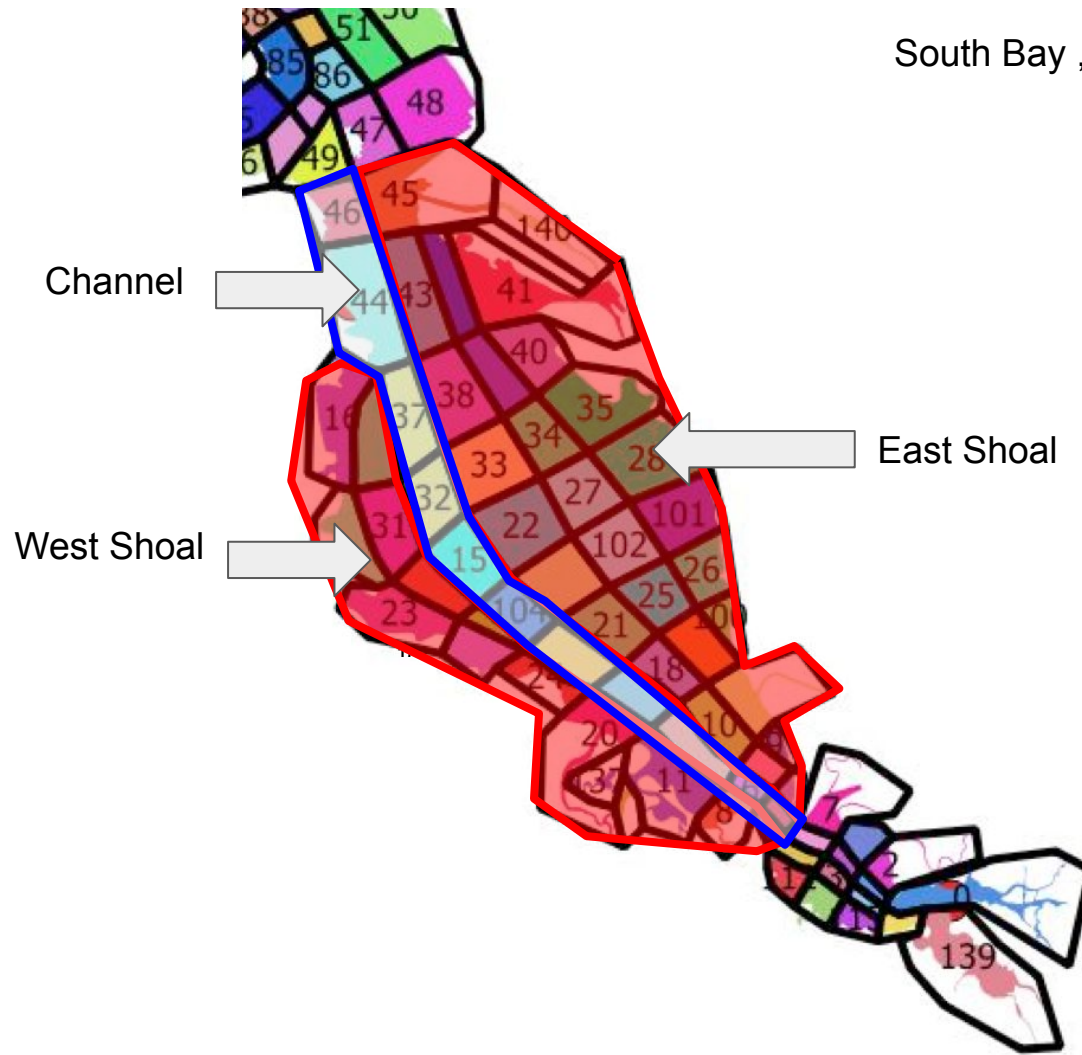
(+9)

- Slide 7...There are interesting transport pathways, in and out for A, B, and C. By keeping T1-T2 and U1-U2 separate for now, I'm wondering if we may be able to qualitatively tell which sources are predominantly moving along flow paths. For example,

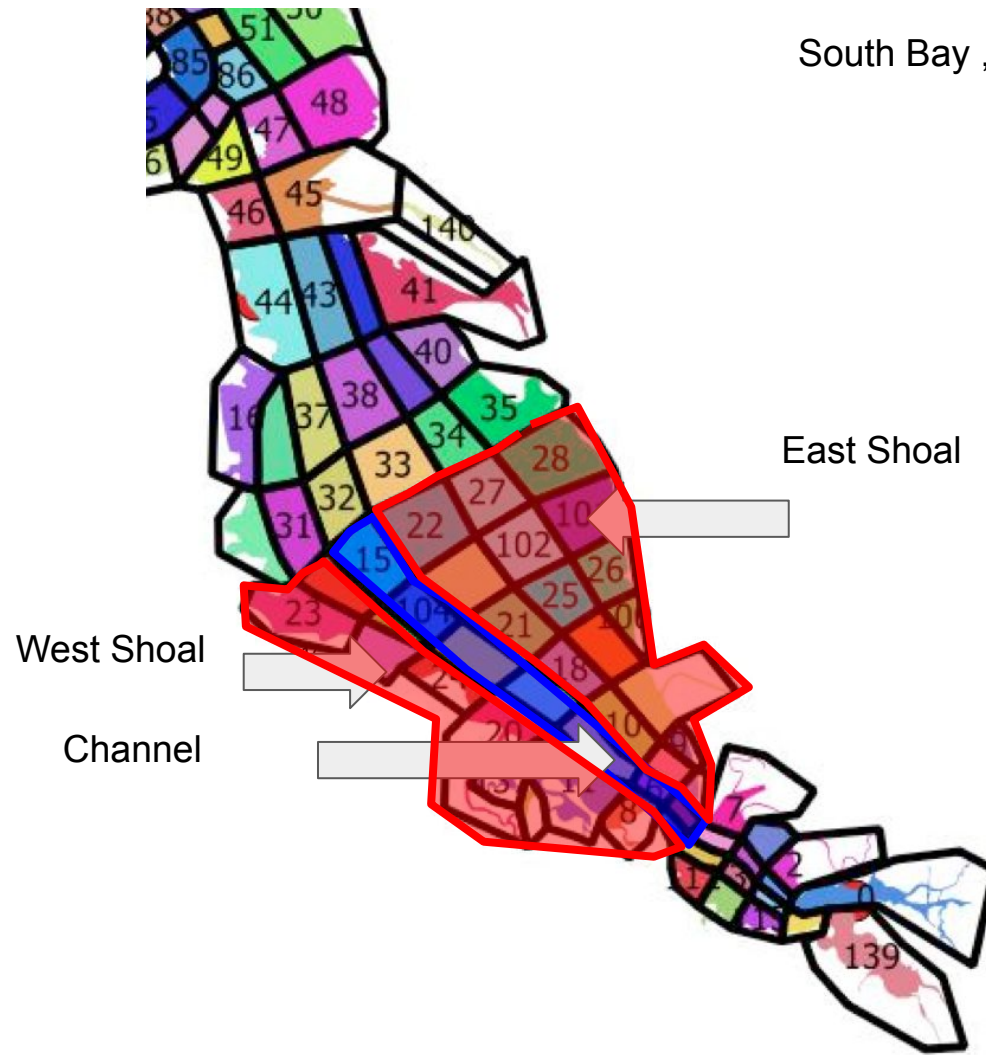
+3.5

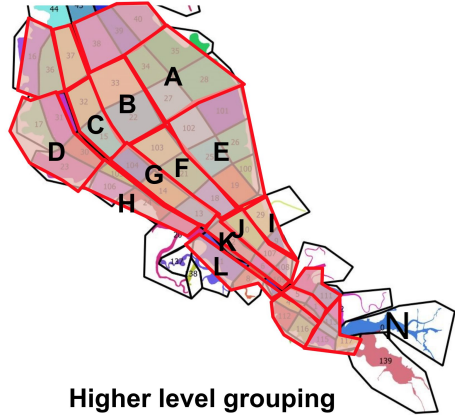
- does a large portion of DIN-fluxA (flux leaving A, transported to north) essentially continue moving north and leaving South Bay; or
- Does it much (based on relative masses) appear to recombine with EBDA and SFPUC DIN and recirculate, and comprise some of the DIN transport south into B
- See individual plots on next page

South Bay , as designated by WB

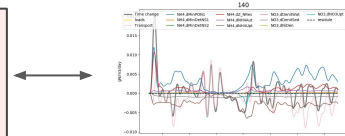
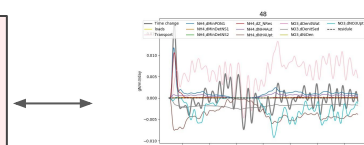
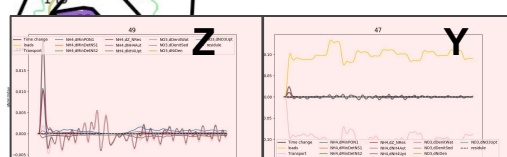


South Bay , as designated by RMP

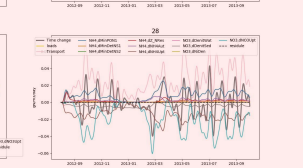
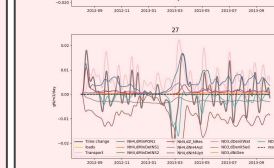
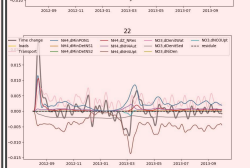
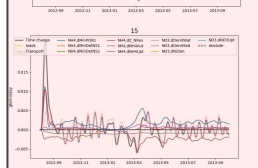
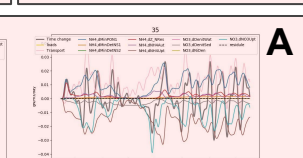
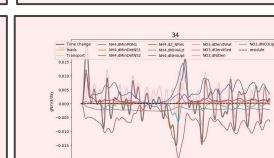
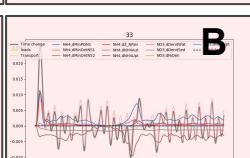
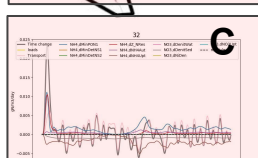
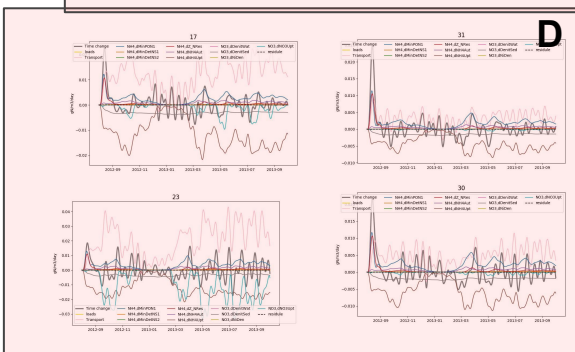
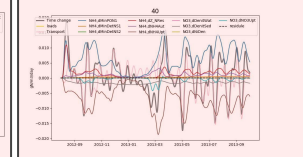
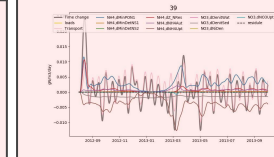
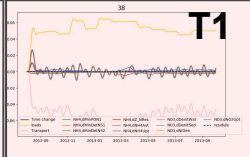
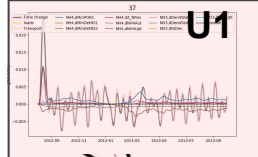
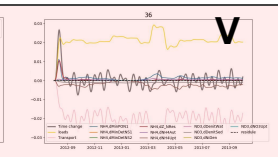
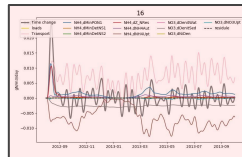
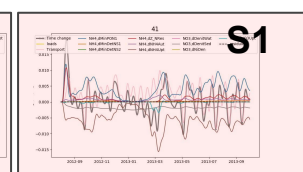
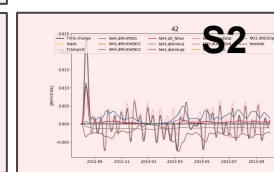
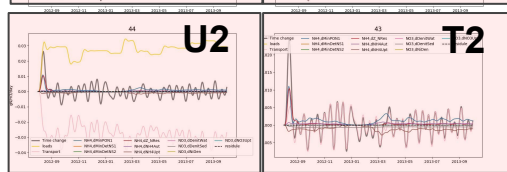
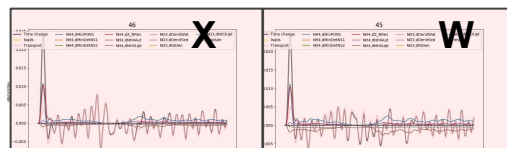




Higher level grouping



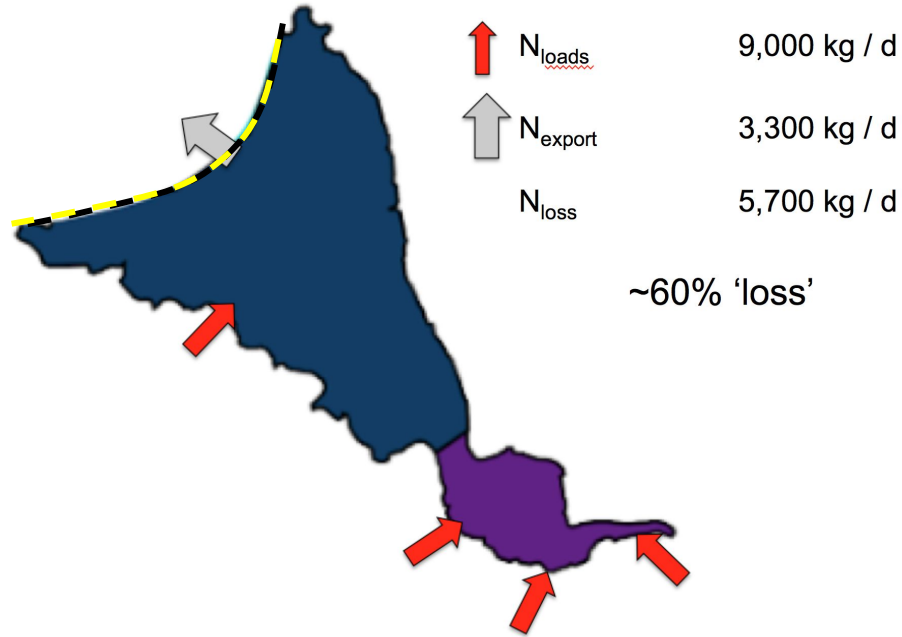
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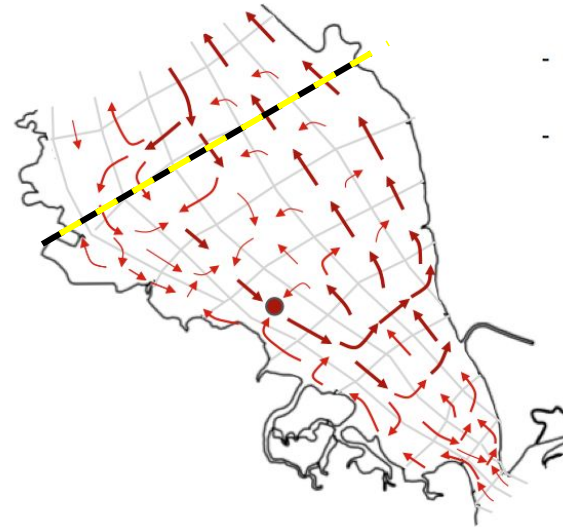
A

Subembayment mass balance (SB+LSB)

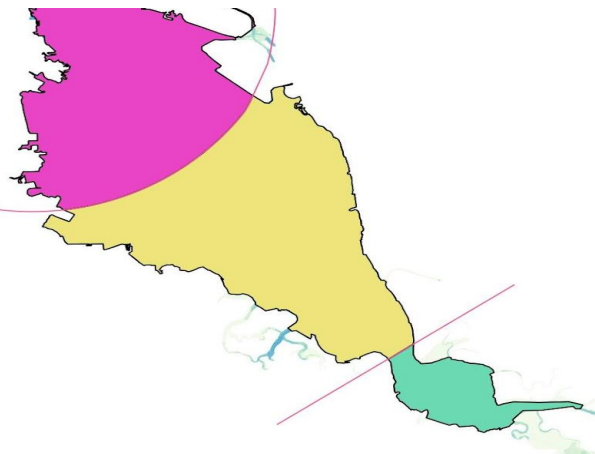
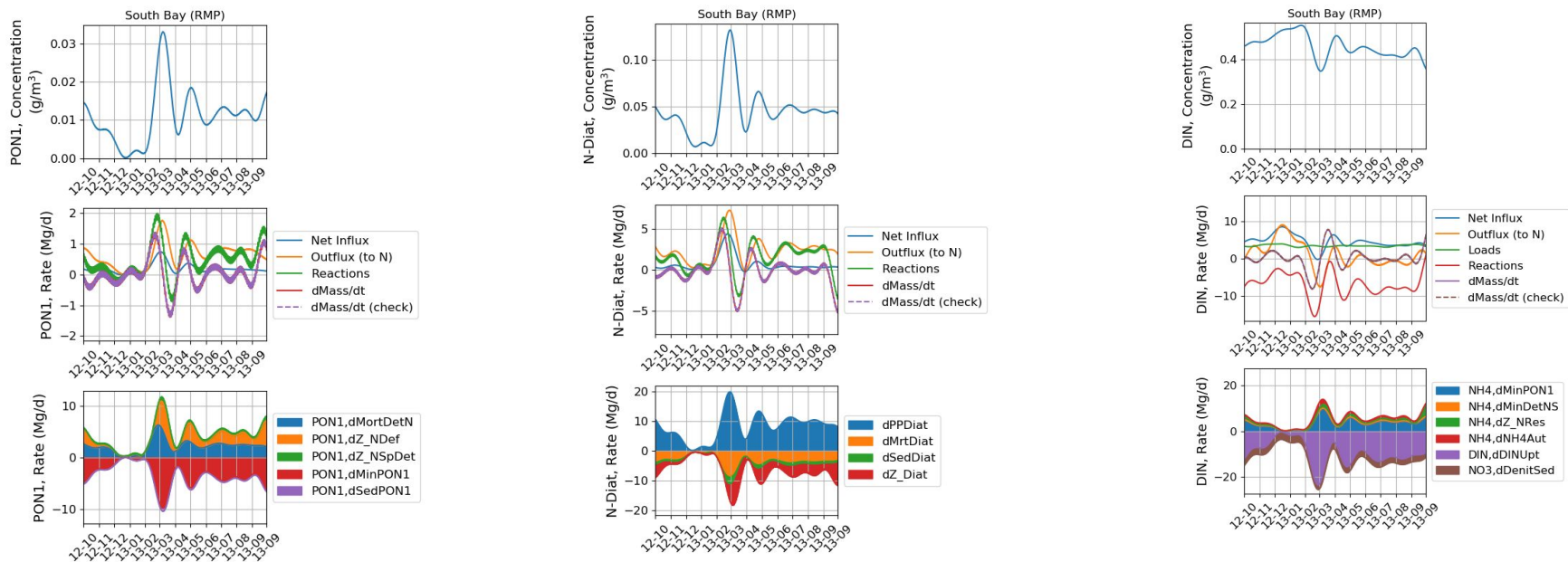
mid-Summer Total N Budget



At higher-resolution...the 'net' export is much more complex.



- Considerably more complex
- Large-magnitude DIN circulation
 - Efficient transport onto the shoal
- Shoal is also where most N removal and most primary production takes place.
- What kind of data do we need?

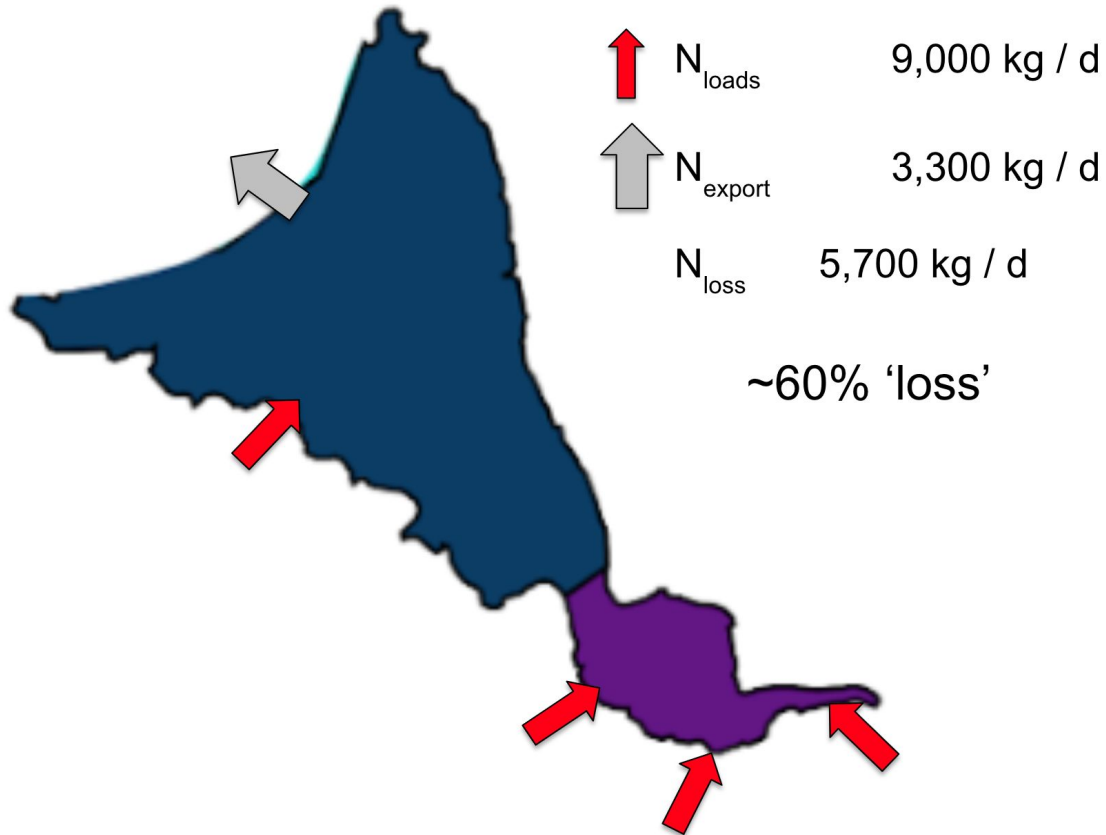


$$N \text{ outflux to N} = 0.8 \text{ (PON)} + 2.5 \text{ (N-diat)} + /- 0 \text{ (DIN)} = 3.3$$

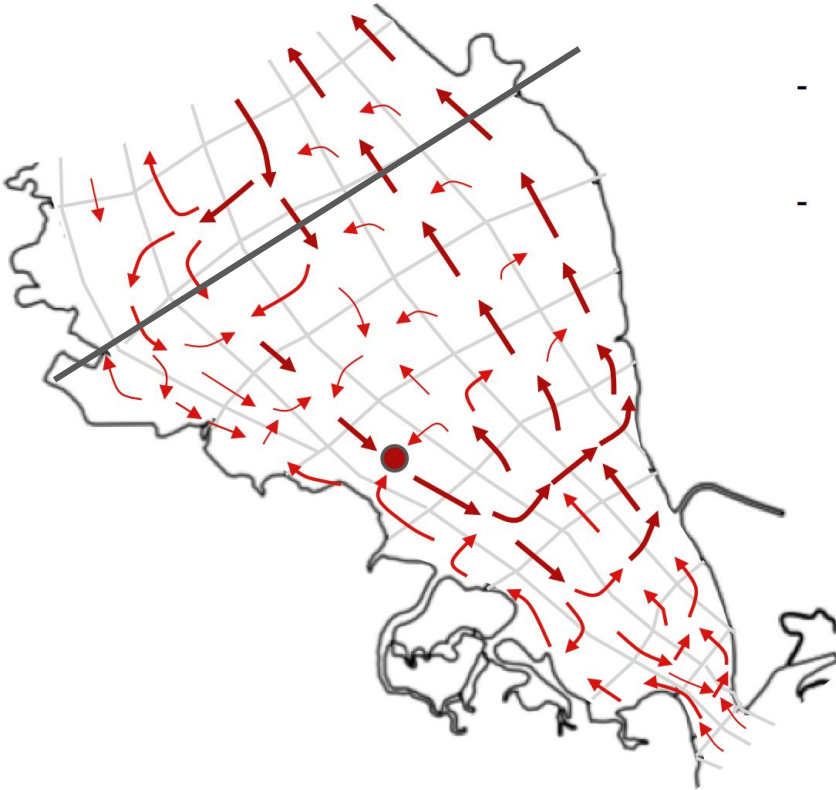
$$\text{Total N inputs WWTPs} = 3 \text{ (DIN)} + 6 \text{ (DIN from LSB)}$$

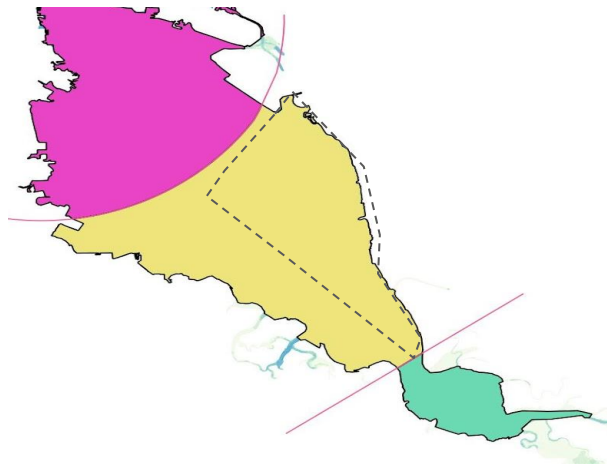
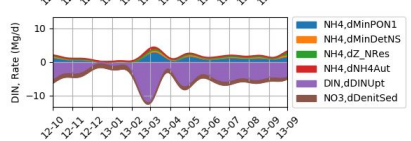
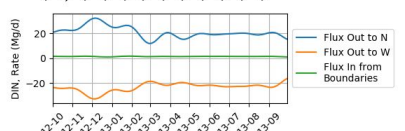
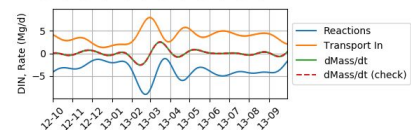
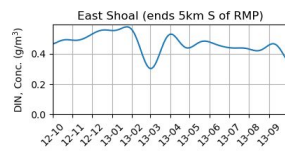
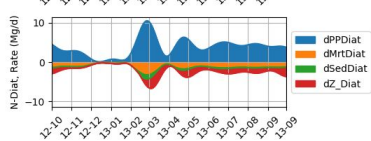
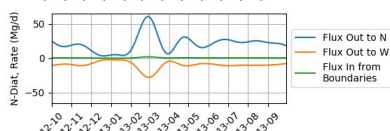
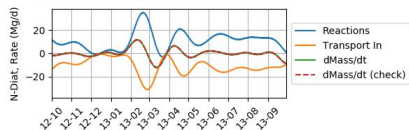
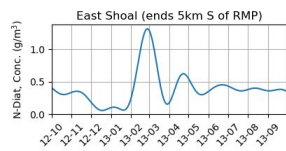
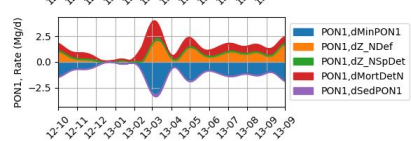
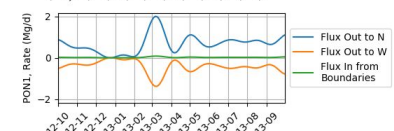
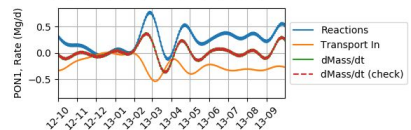
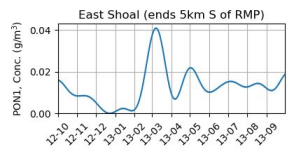
Loss is ~60% summertime

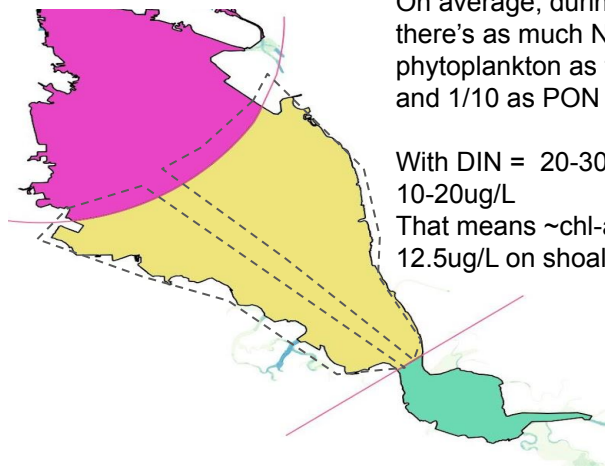
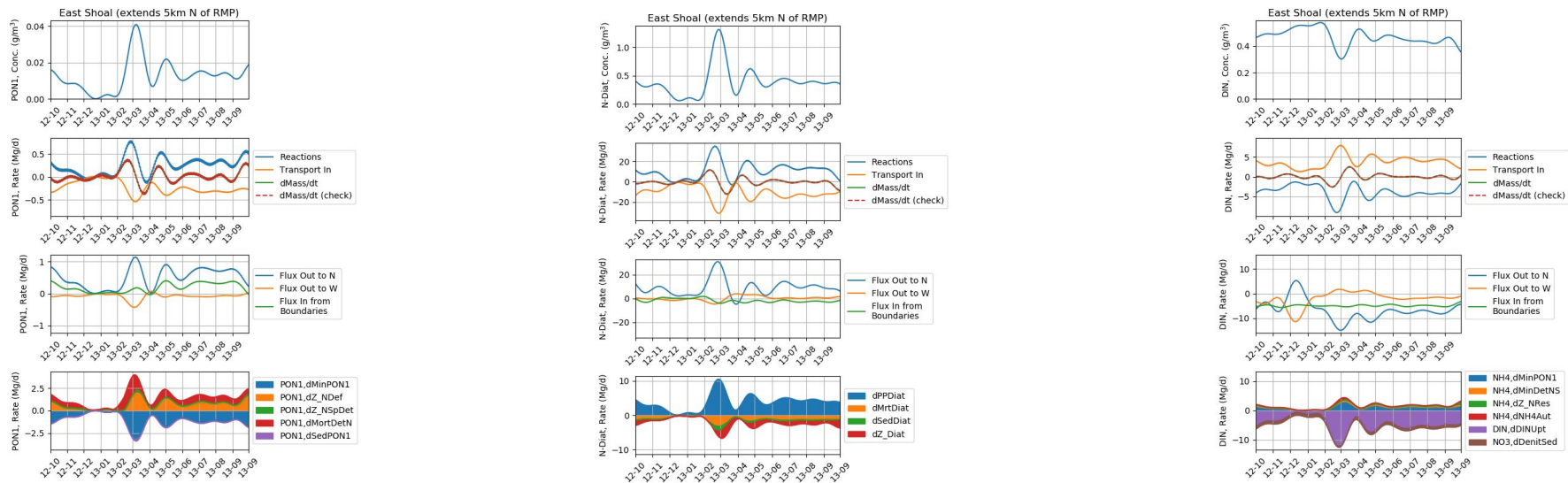
mid-Summer Total N Budget



- Considerably more complex
- Large-magnitude DIN circulation
 - Efficient transport onto the shoal
- Shoal is also where most N removal and most primary production takes place.
- What kind of data do we need?



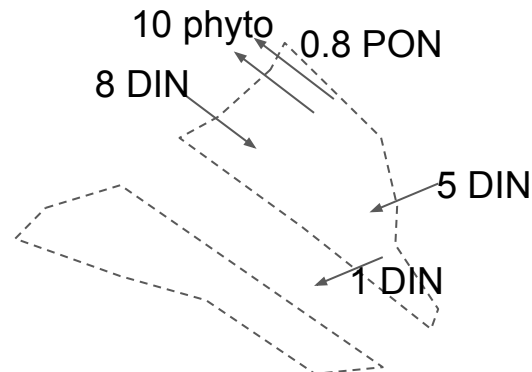


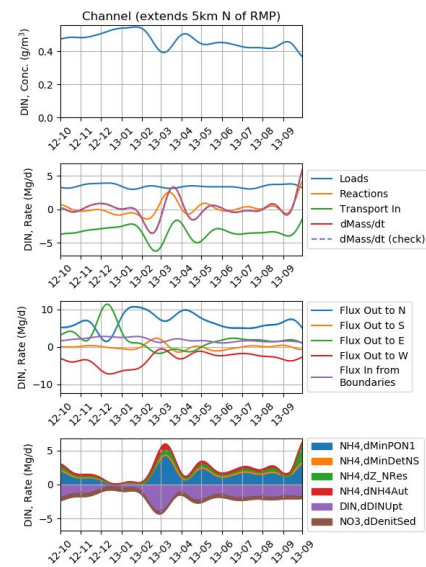
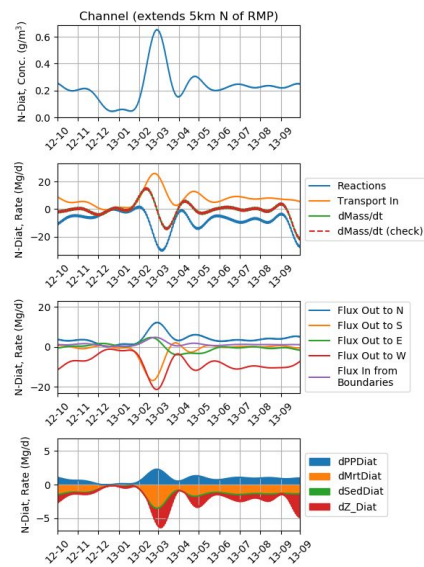
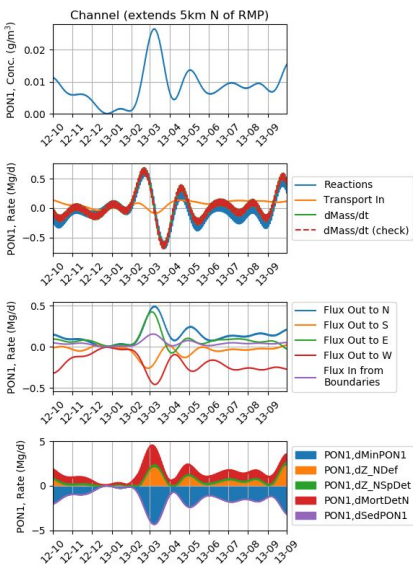


On average, during summer,
there's as much N in
phytoplankton as there is in DIN;
and 1/10 as PON

With DIN = 20-30 μM , and
10-20 $\mu\text{g/L}$
That means $\sim\text{chl-a}$ around
12.5 $\mu\text{g/L}$ on shoal steadily

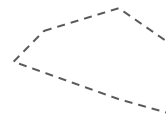
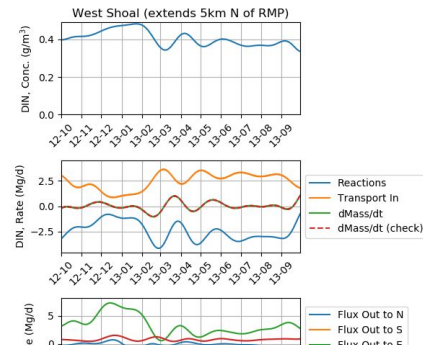
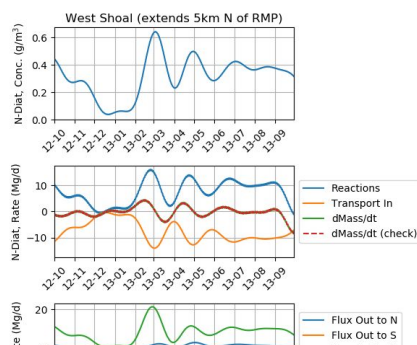
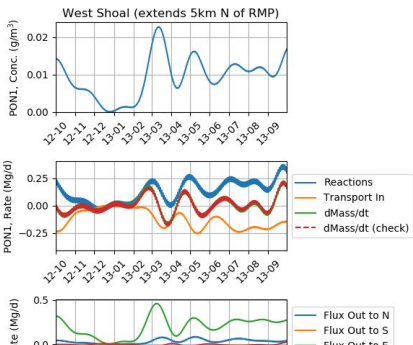
$$\text{Flux out to North} = -8 \text{ (DIN)} + 10 \text{ (phyto)} + 0.8 \text{ PON}$$





West shoal = Exports 10 (n-phyto) (to channel) + -4.5 (from south, input) + 2 (to channel)

Channel = export+ 3 n-phyto to N - 7 phyto from west (n-phyto) (to channel) + + 3 DIN load input) + 5 to N - (to channel)



West shoal = Exports 10 (n-phyto) (to channel) + -4.5 (from south, input) + 2 (to channel)

This doesn't add up, production and export of
N=phyto = 10,

Only 4.5 DIN + 0.5 (boundaries) entering, was the
change in concentration over prior periods
enough to balance?

1 phyto + 0.05 PON

0.8 PON

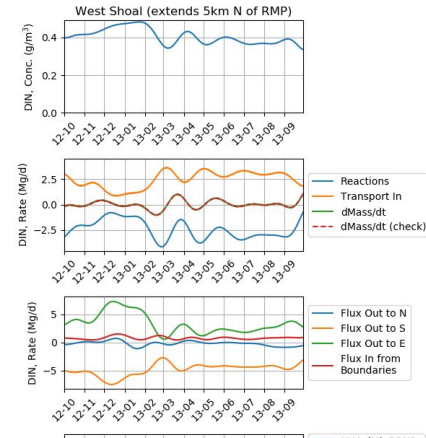
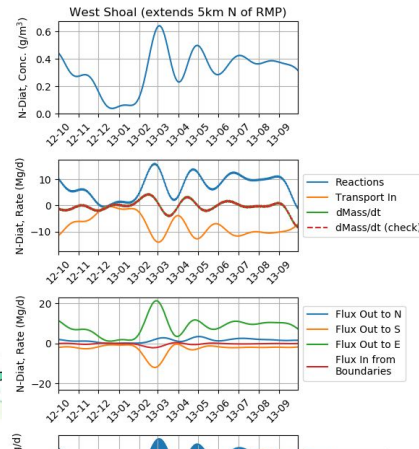
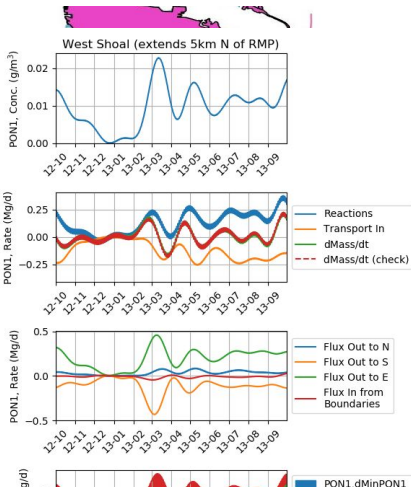
0.5 DIN

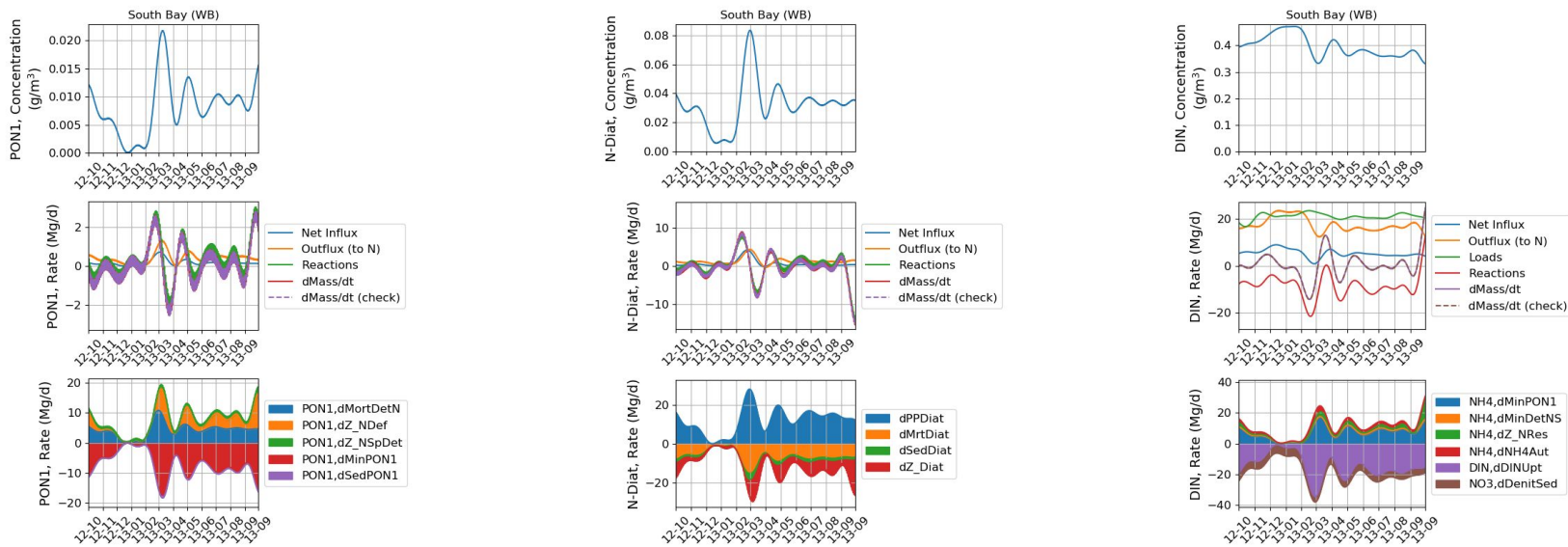
9 phyto+0.25

1.4 DIN

1 phyto + 0.1 PON

4.5 DIN

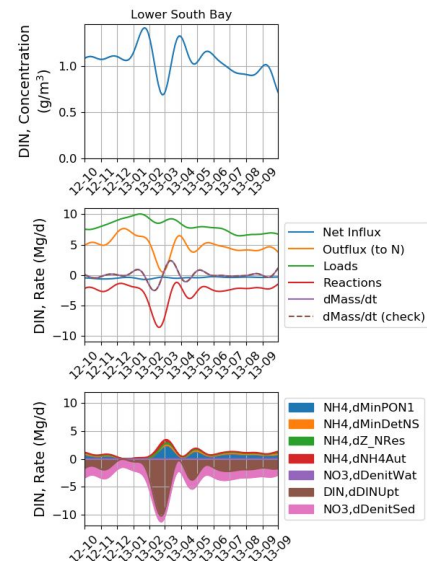
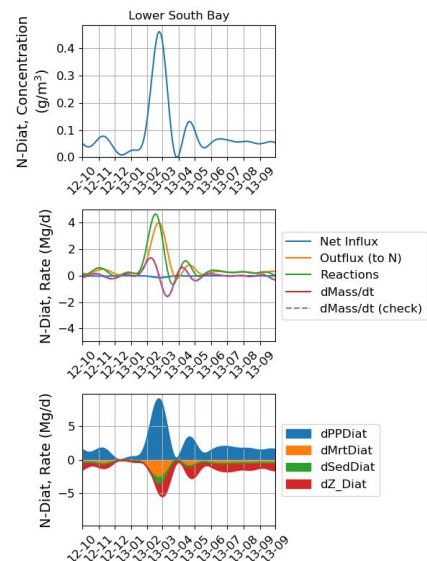
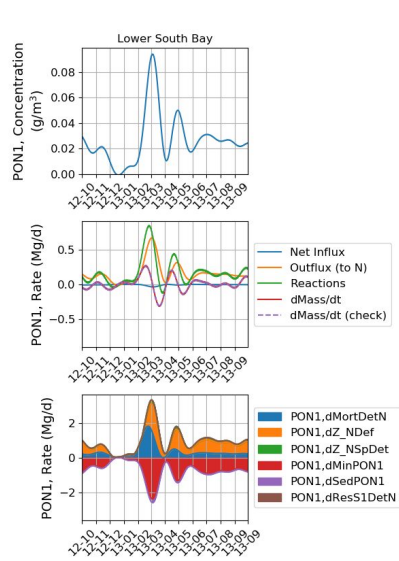




$N \text{ outflux to N} = 15 + 0.25 \text{ PON} + 1 \text{ N-diat}$

Total loads = 20 + 6 (south of dumbarton)

10/26



$$\text{Efflux} = 4.5 (\text{DIN}) + 0.25 (\text{N-phyto}) + 0.25 (\text{PON})$$